

# Madhurima Nath

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madhurima-nath.github.io || [GitHub](#) || [LinkedIn](#) || [Google Scholar](#)

Senior data consultant with a PhD in computational physics, based in New York City, USA. Research in network science and complex systems; industry work in generative AI, NLP, and data engineering.

## POSITIONS

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|--------------------------------------------------------------------------|---------------------|
| <b>Senior Data Consultant</b> , Slalom, Inc., New York, New York, USA    | Sep 2024 – Present  |
| <b>Data Consultant</b> , Slalom, Inc., New York, New York, USA           | Jul 2021 – Aug 2024 |
| <b>Associate Data Consultant</b> , Slalom, Inc., New York, New York, USA | Jan 2020 – Jun 2021 |
| <b>Post-doctoral Research Assistant</b> , Virginia Tech, Virginia, USA   | Feb 2019 – Dec 2019 |
| <b>Instructor, Department of Physics</b> , Virginia Tech, Virginia, USA  | Aug 2018 – Dec 2018 |
| <b>Graduate Research Assistant</b> , Virginia Tech, Virginia, USA        | May 2014 – Dec 2018 |

## EDUCATION

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| <b>Ph.D., Physics (Computational)</b> , Virginia Tech, Virginia, USA                                                                                                                                                                                                                                                                                                                                                                              | Dec 2018 |
| <ul style="list-style-type: none"><li><i>Research Focus</i>: Complex systems analysis, Statistical physics, Network science, Community detection</li><li><i>Dissertation</i>: Application of Network Reliability to Analyse Diffusive Processes on Graph Dynamical Systems</li><li><i>US Patent</i> (US20210286859A1): System, method and computer readable medium for sensitivity of dynamical systems to interaction network topology</li></ul> |          |
| <b>M.S., Physics</b> , Virginia Tech, Virginia, USA                                                                                                                                                                                                                                                                                                                                                                                               | May 2017 |
| <b>M.Sc., Physics</b> , Indian Institute of Technology Delhi, India                                                                                                                                                                                                                                                                                                                                                                               | May 2012 |
| <ul style="list-style-type: none"><li><i>Thesis</i>: Study of Cold Atomic Condensates by Atomic Photon Interactions</li><li><i>Award</i>: Best Master of Science Thesis 2011-2012</li></ul>                                                                                                                                                                                                                                                       |          |
| <b>B.Sc. (Hons.)</b> , Physics, University of Calcutta, India                                                                                                                                                                                                                                                                                                                                                                                     | May 2010 |
| <ul style="list-style-type: none"><li><i>Minors</i>: Mathematics and Chemistry</li></ul>                                                                                                                                                                                                                                                                                                                                                          |          |

## GENERATIVE AI & LARGE LANGUAGE MODEL (LLM) PROJECTS

### Data Modernisation to Databricks (3 months) || AI Engineer || Major US Healthcare Client

- Migrated legacy code streams to Databricks Asset Bundle repositories on GCP under Unity Catalog governance using an AI-automated conversion tool, saving weeks of manual effort.
- Delivered the first phase by ingesting 94 patient and 68 supply chain data objects into Bronze layers, subsequently developing custom Silver/Gold layers to reduce latency from 8-10 hours to near real-time.
- Solely engineered enhancements for this company-wide tool using Claude Code; built a reference-document backend for LLM context ingestion, added BigQuery support, and resolved configuration errors in the generated DAB output.

### Product Catalogue Discovery Assistant (2 months) || Lead Data Scientist || Major US Financial Services Client

- Designed and developed an agentic search (RAG) system on the Databricks Mosaic AI platform, integrating Llama models with LangChain agents to automate product document retrieval for both business clients and product specialists.
- Developed a production-ready POC that reduced query resolution times and extracted precise product matches, outperforming manual keyword searches and successfully passing client evaluation.
- Established robust MLOps pipelines and enterprise governance by configuring MLflow for experiment tracking and versioning, integrating Unity Catalog to securely manage vector databases and implement regulatory compliance guardrails.

### Clinical Data Support Q&A Chatbot (6 weeks) || Data Scientist || Healthcare Technology Startup

- Built a Q&A assistant using Anthropic Claude on AWS Bedrock to summarise complex patient clustering data into clear English for healthcare providers.
- Applied prompt engineering to combine clustering results with structured patient data including symptoms, health conditions, and drug classifications from JSON files.
- Developed a proof of concept for bio-pharmaceutical data extraction that the client adopted for production; worked with designers to improve interface accessibility.

## DATA SCIENCE PROJECTS

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### **NLP for Safety Analytics (5 months) || Lead Data Scientist || Major US Energy Utility**

- Developed and deployed end-to-end NLP pipeline on Azure Databricks to process five years of safety incident reports from multiple sources in a unified data lake.
- Implemented LDA (Latent Dirichlet Allocation) topic modelling and sentiment analysis to extract safety themes and trends from unstructured incident narratives across electricity and gas operations.
- Deployed production solution with automated weekly reports and interactive dashboards enabling executives to filter insights by business unit and monitor emerging themes.
- Solution received executive recognition and was adopted across gas, electric, and storage business units, providing insights that informed safety policy and resource allocation decisions.

### **NLP for Global Procurement Standardisation (8 months) || Data Scientist || Major US Consumer Packaged Goods Company**

- Built text classification pipeline using Random Forest on Azure Machine Learning to automatically classify products into standardised global category templates across Latin American and North American procurement catalogues.
- Collaborated with engineering teams to incorporate MLOps (machine learning operations) best practices including automated pipeline updates and model retraining workflows for production deployment.
- Removed a critical data standardisation bottleneck, accelerating the client's delivery timeline by ~1 year and improving data consistency across global procurement operations.

## DATA ENGINEERING PROJECTS

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### **Data Lake Implementation on Snowflake (12 months) || Data Engineer || Nonprofit Organisation**

- Maintained and upgraded a Snowflake data lake ingesting 2M daily FHIR records across 6 regional health networks, powering analytics for ~5 million residents.
- Built Snowflake task orchestrations and parameterised stored procedures for automated batch ingestion, integrating automated data quality checks and HIPAA-compliant privacy controls.
- Refactored legacy monolithic JavaScript into modular Python and SQL, reducing technical debt and enabling seamless pipeline extension across additional source networks.

### **Network Predictive Maintenance Solution (4 weeks) || Analytics Engineer || Major US Telecom Client**

- Engineered a predictive maintenance pipeline combining 6 radio frequency (RF) metrics with geographical clustering to proactively isolate faulty network infrastructure from user device data.
- Implemented automated Google BigQuery stored procedures processing 90-day performance data across the New York metropolitan area, collaborating with network engineering teams to validate preventive insights.
- Delivered ~US \$8M in projected annual cost savings by reducing ~30K customer support calls and ~6K service visits, with the methodology validated for nationwide scaling.

### **Data Migration to Salesforce (5 months) || Data Engineer || Nonprofit Organisation**

- Migrated 1TB of historical data to Salesforce via custom SQL stored procedures, preserving complex foreign key relationships across interdependent datasets of billions of rows.
- Configured parameterised Azure Data Factory (ADF) pipelines supporting incremental loads, leveraging table partitioning to optimise large-scale dataset performance.
- Built an automated validation framework (including row counts, source-to-destination comparisons, and duplicate detection) to guarantee 100% data integrity during live migrations.

## TECHNICAL EXPERTISE

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- **Programming:** Python, SQL, PySpark, R, Bash || **CI/CD:** GitLab, GitHub, Azure DevOps, BitBucket
- **Distributed Computing:** Databricks, Snowflake || **Cloud Platforms:** AWS, Azure, GCP
- **ML Frameworks:** MLflow, LangChain, TensorFlow (familiar) || **Data Tools:** Delta Lake
- **GenAI Development:** Prompt engineering, multi-agent frameworks, RAG (retrieval-augmented generation)
- **Project Management:** Agile/Scrum (JIRA), cross-functional team leadership, stakeholder communication

## ACADEMIC RESEARCH

### Post-doctoral Research Assistant || Virginia Tech, USA || *Supervisor: Prof. Stephen Eubank* Feb 2019 – Dec 2019

- Developed hybrid computational approach combining Monte Carlo simulations with weak- and strong-coupling perturbative expansions to improve estimation of Moore-Shannon network reliability on graphs, addressing an NP-hard computational problem. [2]
- Extended the framework to heterogeneous satisfiability problems, constructing tight upper and lower bounds on approximation error for mostly monotonic probabilistic satisfiability problems. [2]
- Implemented statistical approaches for community detection in large-scale weighted directed networks, achieving significant improvements over traditional methods. [1]

### Graduate Research Assistant || Virginia Tech, USA || *Supervisor: Prof. Stephen Eubank* May 2014 – Dec 2018

- Applied Moore-Shannon network reliability formalism (traditionally used for electronic circuit analysis) to biological and social network systems, establishing theoretical framework connecting electrical engineering concepts to network epidemiology and complex systems analysis. [GitHub repo](#) [4, 5]
- Developed computational approaches to evaluate Moore-Shannon network reliability formalism using Bernstein basis functions as polynomial basis set, providing convergence guarantees for sequential design and model selection in discrete finite systems. [GitHub repo](#) [2]
- Showed that synthetic network models generated using Exponential Random Graph Models cannot reliably predict epidemic outcomes on empirical contact networks. Analysis of National Longitudinal Study of Adolescent to Adult Health (Add Health) dataset demonstrated that models matching local network statistics can overestimate infection numbers by  $\sim 50\%$ . Applied Birnbaum importance measures to quantify individual edge contributions to epidemic potential, enabling identification of critical connections for targeted mitigation strategies. [5]
- Identified vulnerabilities within global food trade networks using commodity-specific data from UN Comtrade database, analysing where pest and pathogen contamination could cascade through global supply chains. Achieved  $\sim 96\%$  precision in forecasting impact of mitigation strategies under various contagion scenarios. [3]
- Reduced time complexity from  $O(n^2)$  to  $O(n)$  for estimating energy states in interacting magnetic systems (Ising model) by reformulating the problem through network reliability framework. Established equivalence between network reliability and Ising partition function, enabling transfer of statistical physics computational methods to network science applications. [7]
- Developed parallel Markov-chain Monte Carlo scheme for estimating joint density of states in Ising model applications, addressing fundamental limitations of naive sampling approaches that yield poor approximations for partition function estimation. [7]

### Master's Thesis || Indian Institute of Technology Delhi, India || *Supervisor: Prof. Sankalpa Ghosh* Jul 2011 – May 2012

Study of Cold Atomic Condensates by Atom-Photon Interactions [9] || Best Master of Science Thesis 2011-2012

- Developed computational framework in MATLAB for probing quantum many-body states of ultracold atoms in optical lattices using angle-resolved light transmission measurements, demonstrating that confined atoms act as quantum diffraction gratings.
- Calculated dispersive shifts in cavity resonance caused by atomic presence, establishing direct proportionality between resonance shift and atom count in illuminated sites for systematic quantum state characterisation.
- Generated transmission spectra across Mott Insulator and Superfluid phases, establishing a non-destructive approach for probing Fock-space structure in few-body correlated quantum systems.
- Visualisation selected for American Physical Society (APS) Physical Review A [Kaleidoscope](#) (June 2012) based on aesthetic quality of graphics.

## TEACHING EXPERIENCE

### Instructor, Department of Physics, Virginia Tech, USA Fall 2018

- Designed and delivered three undergraduate physics modules: Optics, Thermal Physics, Waves & Sound, to  $\sim 90$  students across all modules.
- Developed all teaching materials including lectures, problem sets, examinations, and assessment criteria.
- Managed course administration including student records and academic standards compliance.
- Achieved above-average student feedback scores for creating respectful learning environments and fair assessment practices.

**Graduate Teaching Assistant, Department of Physics, Virginia Tech, USA****2013–2015**

- Taught introductory physics across four semesters: PHYS 2305/2306 (calculus-based, for physical sciences, mathematics, and engineering students) and PHYS 2205/2206 (algebra-based, for all other disciplines).
- Conducted weekly laboratory sessions and recitations for 10–15 students covering mechanics, electromagnetism, and optics; held office hours open to all enrolled students four times weekly.
- Received positive feedback for fostering inclusive learning environments and providing effective student support.

**SUMMER RESEARCH PROJECTS****Quantum Information Conservation: The No-Hiding Theorem || Completion letter****Jun 2012 – Jul 2012**Harish-Chandra Research Institute, India || *Supervisor*: Prof. Arun Kumar Pati

- Selected for the Visiting Students Programme (VSP) in Physics at a premier research institute funded by the Department of Atomic Energy, Government of India, to conduct research in quantum information theory.
- Studied the No-Hiding Theorem through theoretical analysis and numerical verification, establishing that quantum information cannot be created or destroyed but only redistributed between system and environment; analysed perfect hiding scenarios where information transfers completely to ancilla states without entanglement.
- Quantified information distribution in imperfect hiding cases using correlation measures implemented in MATLAB.

**Renormalisation Group Study of Liénard Systems || Completion letter****May 2011 – Jul 2011**Jadavpur University, India || *Supervisor*: Dr. Dhruba Banerjee

- Applied renormalisation group methods to analyse limit cycle behaviour in nonlinear dynamical systems, working within the framework of Hilbert’s sixteenth problem on limit cycles in polynomial differential equations.
- Derived amplitude equations for generalised Liénard systems using perturbative renormalisation group techniques; determined limit cycle existence and stability by analysing fixed points and eigenvalues of the amplitude equations.
- Validated theoretical predictions through computational analysis in Mathematica, generating phase portraits and limit cycle trajectories across parameter space.

**PROFESSIONAL & COMMUNITY SERVICE****Peer Review**

- NeurIPS (Neural Information Processing Systems): Machine Learning and Physical Sciences 2022, 2024
- ICML (International Conference on Machine Learning): Synergy of Scientific and Machine Learning Modeling 2023
- Physical Review E 2022 – Present

**Talks & Workshops**

- *Implementing Topic Modeling in the Industry*, Women in Machine Learning and Data Science (WiMDS) Aug 2023
- Insights into Data Engineering (Parts 1 & 2), Women Who Code San Francisco [Part 1](#) || [Part 2](#) Jul 2021
- Exploring NLP Fuzzy Matching Algorithms, Women Who Code [YouTube](#), [Medium article](#) Apr 2021
- Statistics in Data Science, Women Who Code Statistics in Data Science Workshop Series [YouTube](#) Mar 2021
- Regression & Predictions, Women Who Code Data Science: Statistics Workshop Series [YouTube](#) [Medium article](#) Feb 2021
- Technical writing, [Medium](#): data engineering, NLP, and topic modelling 2021 – Present

**Community Engagement**

- Industry Ambassador, Women in Network Science (WiNS) Society 2022 – 2025
- Conference Organiser, WiNS Satellite at Networks 2021 (joint Sunbelt/NetSci) 2021
- Graduate Student Panel speaker, APS Conference for Undergraduate Women in Physics, Virginia Tech, USA Jan 2017
- Physics outreach, Virginia Tech, USA: demonstrations on waves, sound, and resonance at local schools 2015 – 2018
- Associate Editor, *Community Change*, Virginia Tech, USA 2017 – 2018

**Environmental Stewardship**

- Leave No Trace 101 Certification (partnership with Urban Connections programme run by the US Forest Service) Jun 2026
- Climate Intelligence: Using Climate Data to Improve Business Decision-Making, University of Reading (Online) Jul 2026

## PUBLICATIONS

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1. Mishra, R., Eubank, S., **Nath, M.**, Amundsen, M., and Adiga, A. (2022, November). [Community Detection Using Moore-Shannon Network Reliability: Application to Food Networks](#), In International Conference on Complex Networks and Their Applications (pp. 271-282). Cham: Springer International Publishing.
2. Eubank, S., **Nath, M.**, Ren, Y., and Adiga, A. (2022). [Perturbative methods for mostly monotonic probabilistic satisfiability problems](#). arXiv preprint arXiv:2206.03550.
3. **Nath, M.**, et. al. (2019). [Using network reliability to understand international food trade dynamics](#). In Complex Networks and Their Applications VII: Volume 1 Proceedings The 7th International Conference on Complex Networks and Their Applications COMPLEX NETWORKS 2018 7 (pp. 524-535). Springer International Publishing.
4. **Nath, M.**, Ren Y., and Eubank, S. (2019). [An approach to structural analysis using Moore-Shannon network reliability](#). In Complex Networks and Their Applications VII: Volume 1 Proceedings The 7th International Conference on Complex Networks and Their Applications COMPLEX NETWORKS 2018 7 (pp. 537-549). Springer International Publishing.
5. **Nath, M.**, Ren, Y., Khorramzadeh, Y., and Eubank, S. (2018). [Determining whether a class of random graphs is consistent with an observed contact network](#). Journal of theoretical biology. 440, 121-132.
6. **Nath, M.**, and Eubank, S. (2018). [Model selection for sequential designs in discrete finite systems using Bernstein kernels](#). arXiv preprint arXiv:1807.06661.
7. Ren, Y., Eubank, S., and **Nath, M.**. (2016). [From network reliability to the Ising model: A parallel scheme for estimating the joint density of states](#). Physical Review E, 94(4), 042125.
8. **M. Nath**, et. al. (2015). [A two-parameter method to characterise the network reliability for diffusive processes](#). In Complex Networks VI: Proceedings of the 6th Workshop on Complex Networks CompleNet 2015 (pp. 139-148). Springer International Publishing.
9. Agarwala, A., **Nath, M.**, Lugani, J., Thyagarajan, K., and Ghosh, G. (2012). [Fock-space exploration by angle resolved transmission through a quantum diffraction grating of cold atoms in an optical lattice](#). Physical Review A, 85(6), 063606.

## CONFERENCE PRESENTATIONS

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- Network reliability: a generic tool to explore diffusive processes on interacting systems, NASA Prebiotic Chemistry and Early Earth Environments (PCE3) Virtual Workshop 2022 Nano- to Cosmic- Studies of Complex Systems, University of Wisconsin Madison, Madison, WI, USA, 20 Oct, 2022.
- Perturbative methods for estimating relative contributions to network reliability, SIAM (Society for Industrial and Applied Mathematics) Workshop on Network Science, Virtual Workshop, 13-15 Sep, 2022.
- Statistical mechanical applications of graph dynamical systems, Condensed Matter Seminar, Department of Physics, Virginia Tech, Blacksburg, VA, USA, 30 Oct, 2017.
- Determining whether a particular contact network is consistent with a network model, 1st North American Social Networks Conference of the International Network for Social Network Analysis, Washington DC, USA, 26-30 Jul, 2017.
- Network reliability: A novel measure to study the effects of network topology on the diffusive dynamics, Symposium for the Society of Young Network Scientists, NetSci 2017, Indianapolis, IN, USA, 19-23 Jun, 2017.
- Network reliability: A measure to study diffusive dynamics on networks, Center for Soft Matter and Biological Physics Symposium 2017, Virginia Tech, Blacksburg, VA, USA, 17-18 May 2017.
- Effects of network structure on propagation of infectious diseases, 33rd Annual Graduate Student Assembly Symposium and Exposition, Virginia Tech, Blacksburg, VA, USA, 29 Mar 2017.
- Renormalisation group approaches for dynamics on irregular networks, American Physical Society (APS) March Meeting 2017, New Orleans, LA, USA, 13-17 Mar, 2017.
- Diffusive dynamics on a network, Southeastern Section of the American Physical Society (SESAPS) Conference 2016, Charlottesville, VA, USA, 9-12 Nov, 2016.
- Effects of network structure on epidemic modeling, Biocomplexity Institute Symposium 2016, Virginia Tech, Blacksburg, VA, USA, 1 Nov, 2016.
- A two-parameter method to characterise the network reliability for diffusive processes, CompleNet 2015, New York City, NY, USA, 25-27 Mar, 2015.
- Four-parameter characterisation of network reliability and analysis of critical point phenomenology, APS (American Physical Society) March Meeting 2015, San Antonio, TX, USA, 2-6 Mar, 2015.